DocGuard — Module 2 Pedagogical & Technical Guide

Understanding the Native PDF & Metadata Forensic Engine (pdf_forensics.py)

1. What is Module 2 Doing? (The Objective in Simple Language)

Imagine a fraudster wants to tamper with an electronic medical bill or bank statement in PDF format. Instead of printing it out, scanning it, or using complex photo manipulation software, they open the PDF in an online editor like **iLovePDF** or **Canva**, click on the total amount *INR 4,000*, type *INR 14,000*, and hit **Save**.

To human eyes reading the document on a screen, the PDF looks completely crisp, sharp, and authentic. There are no pixel blurs, no low-resolution artifacts, and no obvious visual flaws. Because an investigator does not possess the issuer's original master copy, visual inspection alone fails.

Module 2 is the low-level forensic investigator that inspects the file's internal DNA and digital paper trail.

The Core Forensic Mission:
Before a single pixel is ever rendered or passed to deep neural networks, Module 2 executes a deep structural autopsy directly on the carrier file bytes:
• **Physical File Structure (Binary Autopsy):** Uncovers post-issuance saving sessions, hidden revisions, and unauthorized trailing payloads.
• **Timestamp & Authoring Audit (Metadata Forensics):** Decodes ASN.1 internal dates, audits modification gaps, and checks software blacklists.
• **Typographical Uniformity (Font DNA):** Extracts embedded font dictionaries to detect isolated 'Alien Font' text alterations.
• **Calibrated Risk Scoring Brain:** Fuses independent physical indicators into an empirical risk score (0.0 Clean to 1.0 Tampered).

2. How Does Module 2 Achieve This? (The 4-Stage Forensic Pipeline)

Module 2 processes PDF documents through a sequential multi-stage inspection architecture:

• **Stage 1: Binary File Autopsy:** Ingests the raw byte stream in memory. Validates the ISO 32000-1 magic header (%PDF-), scans backwards for %%EOF markers to detect non-destructive incremental updates, verifies cross-reference tables (startxref), and checks for hidden post-EOF payloads.

• **Stage 2: Metadata & Temporal Audit:** Extracts internal /Info and XMP metadata dictionaries. Decodes ASN.1 timestamps (D:YYYYMMDD...) into UTC datetimes, computes the creation-to-modification delta (Δt), flags negative time travel (date spoofing), and screens authoring signatures against blacklists.

• **Stage 3: Typography & Font DNA Audit:** Inspects low-level font descriptors and character distributions via page.get_text('dict'). Measures character ratios per font family to pinpoint isolated 'Alien Font Injections' (e.g. Courier total inside a Helvetica invoice) and flags typographic font explosions.

• **Stage 4: Calibrated Multi-Vector Risk Scoring:** Fuses binary, temporal, authoring, and typographic anomalies into an empirical risk score S_{risk} in [0.0, 1.0]. Assigns categorical verdicts (CLEAN, SUSPICIOUS, HIGH_RISK) accompanied by deterministic, court-grade explanations.

3. Comprehensive Glossary of Technical, Forensic & ML Concepts

Below is an exhaustive explanation of every technical keyword, binary file principle, forensic indicator, and machine learning concept implemented in Module 2.

A. PDF Binary Architecture & ISO 32000-1 Standards

1. ISO 32000-1 Specification

The international standard defining the Portable Document Format (PDF). It governs how objects (text, fonts, vector shapes, bitmap images) are stored, compressed, indexed, and cross-referenced. Deviations from this standard provide deterministic physical evidence of third-party tampering.

2. Magic Byte Header (%PDF-)

The mandatory leading sequence (e.g. '%PDF-1.4') required within the first 1024 bytes of an authentic PDF. DocGuard verifies this header to prevent arbitrary file upload attacks, extension spoofing, or feeding corrupted binaries.

3. Incremental Updates & Multiple %%EOF Markers

The PDF format allows appending changes to the end of an existing file rather than re-writing the entire document. When an attacker opens a PDF in an editor and re-saves it, the editor appends modified objects, a new cross-reference table, and a second **%%EOF (End Of File)** marker.

• **Forensic Signature:** An authentic, automated institutionally-issued invoice has **exactly 1 %%EOF marker**. A document containing 2 or more %%EOF markers physically proves that the document underwent multiple post-issuance saving sessions.

4. Cross-Reference Tables (xref) and startxref Pointers

PDFs are random-access databases of numbered objects. The cross-reference table (xref) indexes the exact byte offset of each object, and 'startxref' tells the reader where the index begins. In multi-session edits, each revision appends a new xref block and startxref pointer.

5. Trailing Hidden Payloads (Post-EOF Injections)

Standard PDF readers terminate parsing upon reaching the final %%EOF delimiter. Any bytes appended past this marker are completely hidden from human viewers. Forgers and attackers exploit this to conceal secondary documents, keys, or malicious scripts. DocGuard scans beyond the terminal %%EOF to expose hidden payloads.

B. Metadata & Temporal Forensic Analysis

1. Internal /Info Dictionary vs. File System Timestamps

Operating system timestamps (e.g. Windows 'Date Modified') change whenever a file is copied, emailed, or downloaded. In contrast, the PDF internal /Info catalog travels inside the document bytes and contains /CreationDate, /ModDate, /Producer, and /Creator, preserving the authentic digital paper trail.

2. ASN.1 Standardized Timestamp Format

PDF dates follow the ASN.1 format: D:YYYYMMDDHHmmSS[+|-]HH'mm'. For example, D:20261030143000+05'30' represents October 30, 2026 at 14:30:00 UTC+5:30. DocGuard decodes these strings into exact Python datetime instances with full timezone awareness.

3. Temporal Modification Gap (Δt)

The mathematical difference between the modification timestamp and the creation timestamp: Δt = ModDate - CreationDate. In automated accounting, an invoice is generated and finalized instantaneously ($\Delta t < 60$s). A gap where Δt exceeds 86,400 seconds (24 hours) demonstrates that the document was modified in a separate session days or months later.

4. Impossible Temporal Inversion (Negative Time Travel)

A condition where ModDate < CreationDate ($\Delta t < -60$s). Because an authentic file cannot be modified prior to its creation, a negative time gap is mathematical proof of metadata forgery, indicating that a fraudster forged dates manually but failed to adjust both fields synchronously.

5. Software Fingerprinting: Whitelisting vs. Blacklisting

• **Enterprise Whitelist:** Automated, unalterable billing engines such as *ReportLab, SAP ERP, Tally, Oracle NetSuite, QuickBooks*.
• **Consumer Blacklist:** Interactive editors designed for manual document alteration: *iLovePDF, Sejda, Canva, Adobe Photoshop, Nitro Pro, Smallpdf*.
An institutional hospital invoice whose internal Producer reads 'iLovePDF' is an immediate, definitive giveaway of human tampering.

C. Typography & Font DNA Forensics

1. Embedded Font Dictionaries

PDF text objects reference font dictionaries that define character glyph shapes, widths, and encodings. Authentic corporate documents use consistent typography defined in institutional document templates.

2. The 'Alien Font Injection' Attack Signature

When an attacker opens a PDF in an editor to alter a specific number, the editor often lacks the exact font embedded by the original billing software. The editor creates a brand-new font dictionary or injects a fallback system font (e.g. Courier or Arial) solely for that modified text box.
• **Forensic Signal:** DocGuard audits text span character counts. If a document is 75% Helvetica but has an isolated 25% block of Courier containing 'Grand Total: INR 14,000', DocGuard flags an Alien Font Insertion and extracts the exact altered text.

3. Typographical Font Explosion

A single-page document featuring 4 or more distinct font families. Professional billing templates adhere to strict typographic guidelines (typically 1 header font and 1 body font). Font explosions indicate composite document splicing.

4. Scanned / Flattened Document Recognition

Identifies PDFs containing zero vector fonts and zero searchable text—documents that consist solely of a raster image. This alerts downstream modules to rely primarily on visual and pixel-level forensic streams (Module 1 ELA and CNN backbones).

D. Mathematical Fusion & Machine Learning Connections

1. Multi-Vector Evidence Fusion

No single forensic signal is completely immune to false positives. However, when independent physical planes (Binary, Metadata, Typography) compound—such as an incremental update + a blacklisted tool + an alien font—the joint probability of innocent coincidence drops to near-zero.

2. Calibrated Empirical Risk Formulation

The risk engine computes an aggregated risk score: $S_{risk} = \min(1.0, \Sigma w_i \cdot 1_{A_i})$ where w_i represents empirical severity weights:

Forensic Anomaly Indicator (A_i)	Weight (w_i)	Forensic Justification & Physical Impact
Incremental Update (N_EOF >= 2)	0.40	Physical proof of a secondary post-creation save session.
Blacklisted Consumer Software	0.35	Document passed through an interactive editor (e.g. iLovePDF).
Alien Font Injection	0.30	Isolated font introduced specifically for altered character sequences.
Temporal Forgery (Mod < Create)	0.35	Impossible timestamp order (conclusive date manipulation).
Post-Creation Gap (Δt > 24h)	0.25	Document re-saved long after the original generation event.
Hidden Trailing Data (> 32 bytes)	0.25	Unreferenced bytes or payloads appended past final %%EOF.
Font Explosion (>= 4 fonts)	0.20	Typographical inconsistency indicating spliced layout elements.
Stripped / Missing Metadata	0.15	Complete absence of standard /Info metadata headers.

3. Connection to Bayesian Priors in Multimodal Learning

In multimodal machine learning, the Metadata Risk Score functions as an **informative Bayesian prior**. When Stream 3 (Decision Fusion Brain) evaluates a document with S_risk = 0.80, the deep vision networks do not begin from an uninformative 50/50 prior; they are mathematically informed that the carrier file is structurally compromised. Conversely, if S_risk = 0.00, the vision network requires significantly stronger pixel-level visual evidence before triggering a false alarm.

4. Explainable AI (XAI) & Judicial Defense

Deep convolutional networks operate as 'black boxes' that output a single probability score without reasoning. In financial fraud investigations and courtrooms, unexplainable predictions are legally inadmissible. Module 2 delivers **100% explainable, deterministic evidence**: every point added to the risk score is accompanied by exact byte offsets, timestamp discrepancies, and extracted text strings.

Summary: How Module 2 Signals Protect the System

Extracted Signal	Layer Inspected	What It Detects	Downstream Consumer
%%EOF Marker Count	Raw Byte Stream	Post-signing modifications, multi-session saves	Risk Engine & Audit Log
Trailing Byte Count	Post-EOF Bytes	Hidden payloads, appended files, malware	Security Alert
ASN.1 Timestamp Delta (Δt)	/Info Dictionary	Human editing sessions, impossible dates	Risk Engine & Timeline
Software Producer Tag	/Producer & /Creator	Consumer editing tools (iLovePDF, Canva)	Risk Engine & Tool Alert
Font Span Character Ratios	Text Span Dictionaries	Alien font injections, altered numbers	Risk Engine & Text Localizer
Calibrated Risk Score (S_risk)	Unified Aggregator	Holistic structural risk index (0.0 to 1.0)	Decision Fusion Brain